DATA ENGINEERING PROJECT BRIEF

Bank Customer Targeting

From raw, messy records to a ranked marketing target list

PLATFORM	LAYERS	DELIVERABLE
Oracle Database Free (SQL)	**Silver (clean) and Gold (business logic)**	**Ranked list of target customers**

The brief

The Marketing team wants a short, ranked list of high-value customers to target with a new campaign. The data sits in four raw tables that contain NULLs, mixed date formats, duplicates and invalid values. Your job is to clean the data (Silver layer), define what a high-value target means in SQL (Gold layer), and deliver a final list that Marketing can act on.

Source tables

Table	Columns	Keys
CUSTOMER_DEMOGRAPHICS	CUST_ID, FIRST_NAME, LAST_NAME, DOB, PROVINCE, ANNUAL_INCOME	PK: CUST_ID
ACCOUNTS	ACCOUNT_ID, CUST_ID, ACCOUNT_TYPE, ACCOUNT_STATUS, OPEN_DATE	PK: ACCOUNT_ID FK: CUST_ID
TRANSACTIONS	TXN_ID, ACCOUNT_ID, TXN_DATE, AMOUNT, MERCHANT_CATEGORY	PK: TXN_ID FK: ACCOUNT_ID
CUSTOMER_CREDIT_SCORE	CUST_ID, SCORE, LAST_UPDATED	FK: CUST_ID

AMOUNT is positive for deposits and negative for withdrawals. SCORE ranges from 300 to 850.

Pipeline at a glance

STEP 1	STEP 2	STEP 3	STEP 4
Clean and explore	**Build the metrics**	**Apply the criteria**	**Present the list**
Silver layer	Gold layer	Gold layer	Analytics output

1 Data Cleaning and Exploration

SILVER LAYER

Prepare trustworthy data before any business logic touches it.

- [ ] **Fix dates**
  Convert the mixed date formats in the `CUSTOMER_DEMOGRAPHICS` table into a standard `DATE` type.
- [ ] **Handle bad data**
  Write a query to identify how many customers have a negative `ANNUAL_INCOME`. Decide whether to `NULL` them out or exclude them from the final analysis.
- [ ] **Check duplicates**
  Ensure `CUSTOMER_CREDIT_SCORE` only has one score per customer (drop duplicates if necessary).

2 The Logic Engine

GOLD LAYER

Define what a "High-Value Target" is. Write a query that joins all four tables and calculates the following metrics for **Active accounts only**.

#	Metric	Definition
1	**Total Deposits**	Sum of all positive transaction amounts per customer.
2	**Total Spend**	Sum of all negative transaction amounts per customer (absolute value).
3	**Credit Score**	The customer's latest credit score.
4	**Account Age**	How many years the account has been open.

3 The Final Target List

GOLD LAYER

The Marketing team only wants customers who meet **ALL** of the following criteria.

Criterion	Requirement
Credit Score	Greater than 650
Total Deposits	Greater than R50,000 over the last 6 months
Travel activity	At least 5 transactions in the 'Travel' category
Account Age	Greater than 2 years

4 The Analytics Output

WHAT MARKETING SEES

Write **one final query** that outputs the raw list of customers who qualify. Include the columns below.

Column	Description
Customer Name	The customer's name.
Province	The customer's province.
Total Deposits	Total deposits for the customer.
Credit Score	The customer's latest credit score.
Ranking	Rank customers within their `PROVINCE` based on `Total Deposits` using a Window Function, so Marketing knows who the top dogs are in each region.

DELIVERABLES CHECKLIST

Step 1: cleaned, typed and de-duplicated data, plus your bad-data query and your decision on negative incomes.

Step 2: one query that joins all four tables and returns the four metrics for Active accounts.

Steps 3 and 4: one final query that applies every criterion and returns the ranked list.

Written notes on any assumptions you made along the way.